

1. a. How the roads are classified in Nepal? And which institutions are responsible for its development and maintenance?

b. Within strategic road network how many highways and feeder roads are there?

Answer:

According to NRS 2070, roads in Nepal are broadly classified into administrative classification and technical classification.

i. Administrative classification:

- It is important for assigning national importance and level of government responsible for overall management and methods of financing.
- According to this classification, roads are classified into:

a. National highway:

- ✓ They are the main roads connecting East to west and North to South of the nation.
- ✓ These serve for long distance travel, provide consistently higher level of service in terms of speed, travel and inter community mobility.
- ✓ They are designate by 'NH' followed by two-digit number.

b. Feeder roads:

- ✓ These roads serve by connecting district headquarters, major economic centers, tourism centers to national highways or other roads.
- ✓ They are designated by letter F followed by 3-digit number.

c. District roads:

- ✓ These are important roads within a district, serving areas of production and market and connecting with main highways.

d. Urban roads:

- ✓ These roads serve within urban municipalities.
- ✓ According to National Urban Road Standard NURS,2076 urban roads are further classified into
 - **Arterial roads:**
 - These roads are meant for thorough traffic on continuous route.
 - Parking, loading and unloading activities are generally restricted and regulated.
 - Pedestrians are allowed to cross only at the intersections or at designated crossing.
 - **Sub-arterial roads:**
 - These roads are somewhat lower level of travel mobility than arterial roads.
 - Parking, loading and unloading are restricted and regulated.
 - Pedestrian are allowed to cross only at the intersections or at designated crossing.
 - **Collector roads:**
 - It is intended for collecting and distributing the traffic to and from local roads and also for providing access to sub-arterial or arterial roads.
 - They may be located in industrial, residential or business areas.
 - **Local roads:**
 - It provided access to residence, business or other property.

- Such road doesn't carry large volume of traffic.

ii. Technical/ Functional classification:

- For assigning various geometric and technical parameters for design, roads are categorized into classes as follows:

Class I:

- ✓ They are highest standard roads with divided carriageway and access control with ADT 20,000 PCU or more in 20 years perspective period.
- ✓ Design speed adopted for design of this class roads in plain terrain is 120 kmph.

Class II:

- ✓ These are the roads with ADT of 5000-20000 PCU in 20 years perspective period.
- ✓ Design speed adopted for the design of this class of roads in plain terrain is 100 kmph.

Class III:

- ✓ They are the roads with ADT of 2000-5000 PCU in 20 years perspective period.
- ✓ Design speed adopted for design of this class of roads in plain terrain is 60 kmph.

Class IV:

- ✓ These are the roads with ADT of less than 2000 PCU in 20 years perspective period.
- ✓ Design speed adopted for design of this class of roads in plain terrain is 60 kmph.

Roads in Nepal can also be classified into two major categories:

i. Strategic Road Network (SRN)

- a. National Highways
- b. Feeder Roads

Developed and maintained mainly by the Department of Roads (DoR).

ii. Local Road Network (LRN)

- a. District Roads
- b. Urban Roads
- c. Village/Rural Roads

Developed and maintained by Provincial Governments, Municipalities, and Rural Municipalities, with technical support from the Department of Local Infrastructure (DoLI).

The Road Board Nepal (RBN) finances routine and periodic road maintenance through the Road Maintenance Fund.

b. According to the latest records of the Department of Roads, the number of national highways and feeder roads in strategic road network are:

- ✓ National Highways: 80
- ✓ Feeder Roads: 208

2. **a. Calculate the stopping sight distance for a road for which the design speed is 50 kmph. Assume that the roadway is level, the coefficient of friction between the road surface and tires is 0.4 and the reaction time of the driver is 3 sec.**

b. What are the elements of highway cross section? Describe them with a sketch.

Answer:

Given,

Design speed, $V = 50 \text{ km/h}$

Coefficient of longitudinal friction, $f = 0.4$

Reaction time of driver, $t = 3$ s
Road is level.

$$SSD = \text{Lag Distance} + \text{Braking Distance}$$

where ,

$$\text{Lag Distance} = 0.278 \times V \times t$$

$$\text{Braking Distance} = \frac{V^2}{254 f}$$

$$\text{Lag distance} = 0.278 \times 50 \times 3 = 0.278 \times 50 \times 3 = 41.7 \text{ m}$$

$$\text{Braking distance} = \frac{50^2}{254 \times 0.4} = \frac{2500}{101.6} = 24.61 \text{ m}$$

$$SSD = 41.7 + 24.61$$

$$SSD = 66.31 \text{ m}$$

The Stopping Sight Distance (SSD) is:

$$66.3 \text{ m} \approx 67 \text{ m}$$

b. Highway cross-section is the section of road cut perpendicular to the centerline axis of the road. The elements of highway cross-section influence the life of the pavement as well as riding comfort and safety. It consists of many parts such as carriageway, shoulder, median strips, traffic separators, side drain, side slope, catch drains etc.

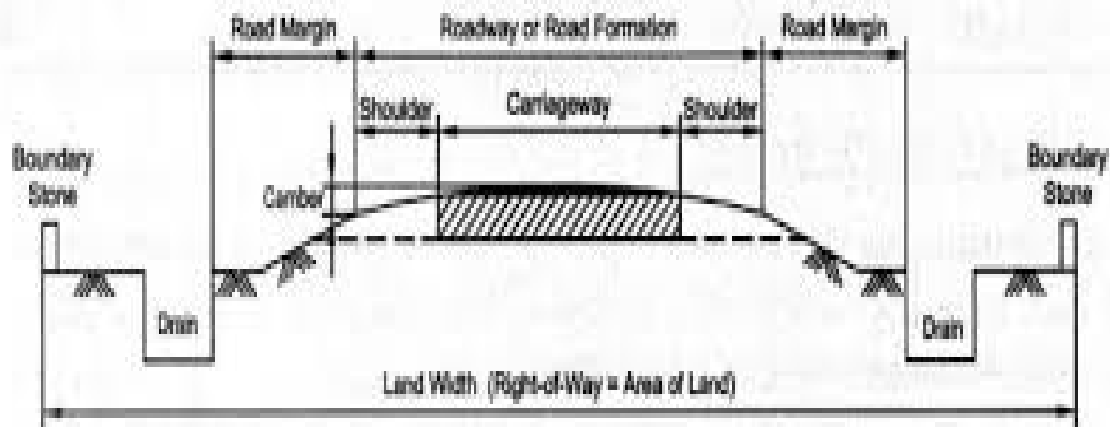


FIG. 1 CROSS-SECTION DETAILS OF A ROAD

i. Carriageway:

- It is the strip of road meant for vehicular traffic movement also known as pavement width.
- Width of carriageway depends upon the width of traffic lanes and number of lanes.
- According to NRS 2070, The minimum lane width for a single lane road is 3.75 m.
- Two lane road is provided with minimum of 3.5 m for each lane.

ii. Shoulder:

- They are the strips provided along the road edge and are intended for the accumulation of stopped vehicles.
- Shoulders serve as emergency lane for vehicles and also provide the lateral support for base and surface courses.
- According to NRS 2070, the width of shoulders on either side of carriageway shall be at least 0.75m.
- It is desirable that color and texture of shoulders be different from those of the carriageway.

iii. Roadway width:

- The sum of width of pavements or carriageway including separators and shoulders is called width of formation or roadway width.
- It is the top width of highway embankment or bottom width of highway cutting excluding the side drains.

iv. Side slope:

- According to NRS 2070, side slopes of embankment and cuttings depend upon the type of fill/ cut materials and height/depth of fill or cut.
- If natural cross slope of the ground is more than 1:5 then the ground should be cut with more than 2 m wide horizontal steps.

v. Lay-bys, bus bays and bus stops:

- Lay-bys are the intermittent shoulders sufficiently wide and long provided to meet the important function of shoulder where the continuous shoulder on either side cannot be provided from economic considerations.
- Bus-bays shouldn't be located too close near the intersections. It is desirable that they are located at least 40 m from the intersection on either side.
- Bus stops should be placed adjacent to the bus linear line of travel so that it doesn't need to pull over to left. It must be provided at the spacings of 200-00 m.

vi. Right of way:

- It is the width of land to be acquired for the road along its alignment. It must be adequate to accommodate all the cross-sectional elements of highway and reasonably provided for future development.
- According to NRS2070,
- ROW requirements for national highways= 50 m (25 m on either side of the road centerline)
Feeder roads = 30 m (15 m on either side of road centerline)
District roads= 20 m (10 m on either side of road centerline)

3. a. What are the different components of hill road drainage system? Make a sketch for a typical drainage system of hill road.

b. Describe the difference between coal and tar. Explain the tests on bituminous materials: penetration test and viscosity test.

Answer:

A hill road drainage system is designed to intercept and remove water from over, under, and the vicinity of the road surface to prevent the softening of the subgrade, loss of structural strength, and water-induced landslides. In mountainous regions like Nepal, effective drainage is critical for protecting the infrastructure from torrential monsoon rains and unstable geological conditions.

Components of hill road drainage system:

The system is categorized into two primary types: surface and subsurface drainage.

i. Surface drainage components:

These components are responsible for removing and diverting water from the road surface and adjoining slopes:

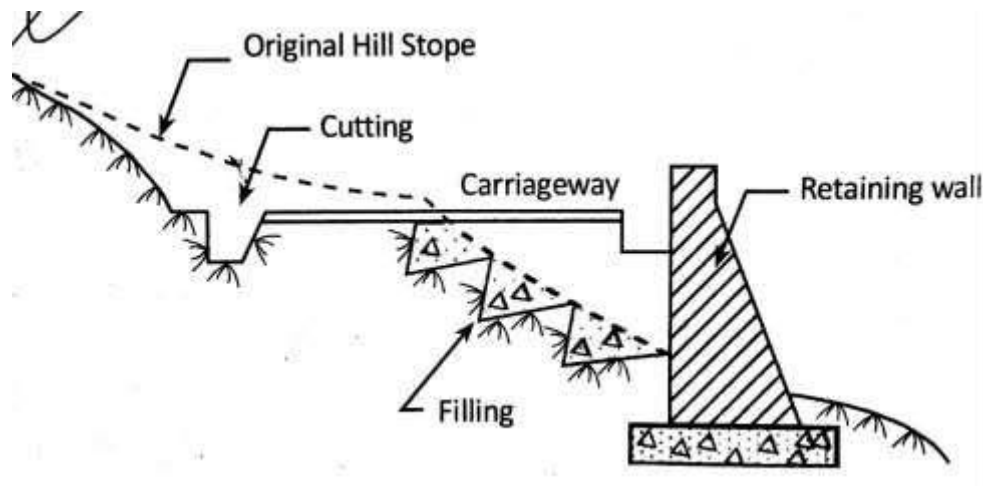
- **Side Drains:** These are open channels provided along the edges of the road (typically on the hill side) to collect runoff from the pavement and cut slopes.
- **Catch-water Drains:** Located on the uphill side of the road, these drains intercept surface runoff from the mountain slopes before it reaches the roadway, diverting it into natural watercourses.

- **Chutes or Cascades:** Stepped structures used to safely convey collected water from higher-level catch drains down steep slopes to lower-level drains or outfalls.
- **Cross-Drainage (CD) Structures:** These allow water from natural streams to pass safely across the road alignment. They include:
 - **Culverts:** For waterways up to 6 m (pipe, box, slab, or arch culverts).
 - **Bridges:** For linear waterways greater than 6 m.
 - **Causeways:** Provided on less important roads where the depth of flow does not exceed 1.5 m.
 - **Aqueducts:** Conduits that drain water across the road from above, particularly useful in hilly areas where culverts are not feasible.

ii. Subsurface drainage components:

These are designed to lower the groundwater table (GWT) and remove seepage from beneath the pavement:

- **Longitudinal Drains:** Trenches with perforated pipes and filter materials installed parallel to the road to keep the GWT at least 1.2 m below the subgrade.
- **Transverse Drains:** Installed across the road to intercept groundwater in relatively impermeable soils.
- **Interceptor Drains:** Constructed uphill to intercept seepage flow before it reaches the road subgrade.
- **Horizontal Drains:** Drilled into the hill slope to relieve excess pore water pressure and stabilize the slope.

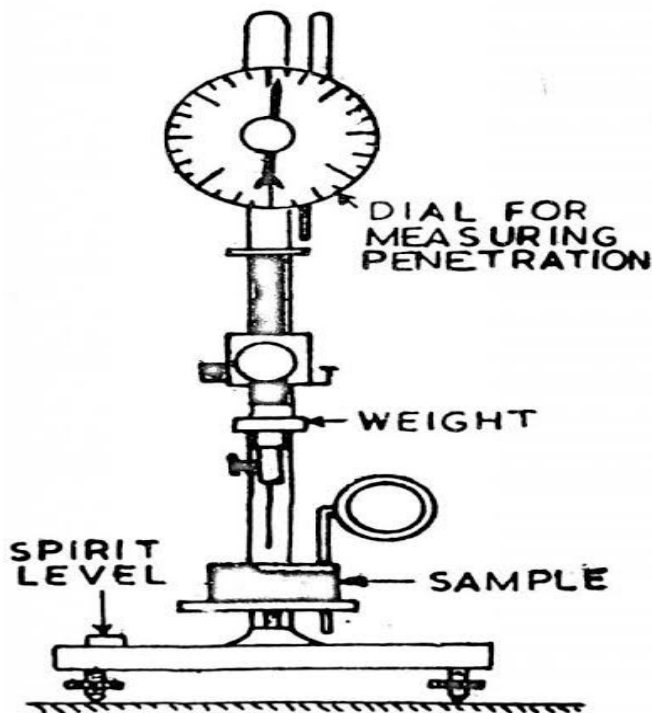


b. Both tar and bitumen are the bituminous road binders used in pavement construction works. The difference between tar and bitumen is described below:

Tar	Bitumen
Tar is the product of destructive distillation of wood or coal in the absence of air.	Bitumen is the product of fractional distillation of crude oil.
It has better affinity in the presence of water.	It has lesser affinity in the presence of water.
Tar has poor weather resistive property.	Bitumen has better weather resistive property.
It is more temperature susceptible.	It is less temperature susceptible.
Tar contains more free carbon.	Bitumen contains less free carbon.
The viscosity varies greatly with temperature.	The viscosity is not susceptible to variation of temperature.

Penetration test:

- Penetration is the measurement of hardness or consistency of bituminous material.
- It measures the vertical distance traversed by the point of a standard needle to the bituminous material under specific condition of load, time and temperature.
- The distance is measured in one tenth of millimeter.



Apparatus Required:

- i. A penetrometer consisting of needle with total weight of 100 gm and device with locking and releasing needle in any position.
- ii. A graduated dial gauge to read penetration value.
- iii. Penetration values can be read with this dial up to 0.1 mm.
- iv. Thermometer readable up to 0.2°C.
- v. Time measuring device with an accuracy of ± 0.1 sec.

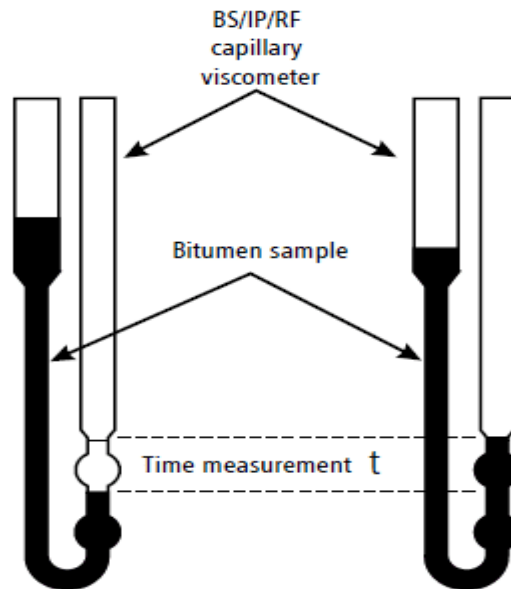
Procedures:

- i. Soften the bitumen to pouring consistency, stir well and pour into the test containers. Keep the depth of bitumen at least 15 mm more than the expected penetrated in container.
- ii. Place the sample containers in water bath at a temperature of 25°C for one hour.
- iii. At the end of one hour take the sample out from the water bath and bring the needle in contact with the surface of bitumen sample.
- iv. Set the reading of the dial gauge at zero, when the needle is in contact with surface of the sample.
- v. Release the needle and allow the needle to penetrate for 5 seconds and note the final reading.
- vi. Take three measurements. The mean value of three measurements is reported as a penetration value.

Penetration test is commonly adopted test on bitumen to grade the material in terms of its hardness.

Viscosity test:

- Viscosity of a fluid is the property by virtue of which it offers resistance to flow.
- Higher the viscosity, the slower will be the movement of liquid.
- The test helps to determine the viscosity of bituminous materials.



Apparatus Required:

- i. An orifice of 10 mm size
- ii. Sample collector
- iii. Thermometer

Procedure:

The time in second for 50 cc of sample to flow through the orifice of 10 mm at the specified temperature of 35, 40, 45 or 55°C.

4. **a. What are the objective and scope of traffic engineering? Explain.**
b. What are the different cause of traffic accidents? Discuss briefly and explain various measures that may take to prevent accidents.

Answer:

a. Traffic engineering is a branch of engineering that covers the road range of engineering applications with a focus on the safety of the public, the efficient use of transportation resources, and the mobility of people and goods. The objectives of traffic engineering are explained below:

- i. To ensure road safety by reducing traffic accidents.
- ii. To maintain efficient traffic movement by minimizing traffic congestion and improving traffic flow and travel speed.
- iii. To increase the road capacity and maximizing the efficient use of existing road infrastructure.
- iv. To reduce travel time and cost while shorten journey times.
- v. To promote environmental sustainability by reducing air pollution, noise pollution and greenhouse gas emissions.

Scope of traffic engineering:

- i. Traffic studies:

It deals with:

- ✓ Traffic volume survey
- ✓ Speed study
- ✓ Origin and destination survey
- ✓ Accident study
- ✓ Pedestrian study

- ii. Traffic control:

It deals with:

- ✓ Traffic signs
- ✓ Road markings:
- ✓ Traffic signals
- ✓ Traffic islands

- iii. Intersection design:
It deals with:
 - ✓ Leveled intersections
 - ✓ Grade separated intersections
 - ✓ Signal timings
- iv. Parking Management:
It deals with:
 - ✓ On street parking
 - ✓ Off street parking
 - ✓ Parking regulations
- v. Traffic regulations and enforcement:
It deals with:
 - ✓ Speed limits
 - ✓ Lane control
 - ✓ One-way streets
 - ✓ Vehicle restrictions

b. Traffic accidents, particularly in congested urban areas with mixed traffic, result from a complex interplay of human, structural, and environmental factors.

- i. **Human Factors (Drivers and Pedestrians):** Over-speeding and reckless driving are identified as the leading causes of accidents, often resulting in severe collisions on narrow road networks. The consumption of alcohol (drunk driving) and distracted driving significantly impair a driver's vision and conscious operation. Additionally, inexperience, fatigue, and the violation of traffic signals contribute to the risk. For pedestrians, jaywalking and the failure to follow crossing rules are prominent causes of fatalities.
- ii. **Road Infrastructure and Design:** Poor Road surface conditions, including potholes, ruts, and corrugations frequently invite accidents. Defective geometric designs, such as inadequate sight distance, insufficient pavement width or shoulders, and improper lighting, further exacerbate safety risks.
- iii. **Environmental and Regulatory Factors:** Unfavorable weather conditions like fog, snow, rainfall, and dust restrict normal visibility, making driving maneuvers unsafe. Furthermore, weak enforcement of traffic laws and improper parking practices contribute to hazardous road environments.

To mitigate the frequency and severity of traffic accidents, a comprehensive strategy involving engineering, enforcement, and education is required:

i. **Engineering Measures:**

- ✓ **Infrastructure Improvement:** Narrow roads should be widened where possible, and surface irregularities like potholes must be repaired promptly.
- ✓ **Geometric Design Correction:** Elements such as horizontal and vertical alignments, sight distances, and pavement characteristics (like skid resistance) must be regularly checked and corrected to meet design standards.
- ✓ **Traffic Facilities:** Proper Road signs, signals, and markings should be installed for clear guidance.
- ✓ **Conflict Reduction:** Constructing pedestrian overpasses, footpaths, and by-passes helps separate through-traffic from local traffic and pedestrians. In high-density areas, grade-separated intersections (flyovers) should be designed to eliminate crossing delays and conflicts.

ii. **Enforcement Measures:**

- ✓ **Speed Control:** Strict enforcement of speed limits through surprise spot-checks and legal actions is necessary to develop correct driving habits.
- ✓ **Licensing and Testing:** Driver licensing procedures should be strengthened with rigorous skill tests during both issuance and renewal.
- ✓ **Vehicle Integrity:** Authorities should levy heavy penalties on defective vehicles and ensure periodic maintenance of critical systems like brakes and steering.

- ✓ **Driver Health:** Regular vision and reaction time checks for drivers should be conducted at prescribed intervals.

iii. **Education Measures:**

- ✓ **User Awareness:** Regular road safety awareness campaigns should be conducted to educate drivers, passengers, and pedestrians on road rules and correct crossing manners.
- ✓ **Training:** Access to professional driving training facilities should be improved to ensure operators possess necessary technical skills.

5. **a. What is ESWL? Explain the concept in the determination of equivalent load.**

b. Distinguish between the full grouted and semi-grouted bituminous macadam.

Answer:

a.

- Equivalent Single Wheel Load (ESWL) is the load of a single wheel that produces the same maximum stress, strain, or deflection at a specified depth in the pavement as a wheel assembly carrying multiple wheels (dual or dual-tandem wheels).
- ESWL converts a multiple-wheel axle load into an equivalent single-wheel load that has the same effect on the pavement.
- It simplifies pavement design calculations.
- The ESWL depends on:
 - ✓ Wheel load
 - ✓ Spacing between wheels
 - ✓ Depth below the pavement surface
 - ✓ Contact area (tire pressure)
- **Principle:**
 - i. At a selected depth below the pavement, determine the vertical stress caused by the multiple-wheel assembly.
 - ii. Find the magnitude of a single-wheel load that produces the same vertical stress or deflection at that depth.
 - iii. This single-wheel load is called the Equivalent Single Wheel Load (ESWL).

b. The difference between the full grouted and semi-grouted bituminous macadam are:

Full grouted macadam	Semi-grouted macadam
i. All the voids in the open-graded bituminous macadam are completely filled with cement grout.	i. Only part of the voids is filled with cement grout.
ii. It produces a dense, rigid and impermeable pavement layer.	ii. It produces a semi-rigid pavement with some remaining voids.
iii. Full grouted macadam has higher strength and load-carrying capacity.	iii. Semi-grouted macadam has moderate strength and load-carrying capacity.
iv. It is more resistant to rutting, deformation and fuel spillage.	iv. It is less resistant to rutting and heavy traffic than full grouted macadam.
v. Construction cost is higher due to complete grouting.	v. Construction cost is lower because less grout is used.

6. **a. Discuss in brief types and methods of maintenance road pavement.**
b. Write down the urgency and importance of highway maintenance. Classify the inspection procedures for the same.

Answer:

a. Highway Road maintenance is the task of preserving, repairing, and restoring a system of roadways to keep their condition as normal as possible and to ensure a long service life. It involves strategic planning, programming, and field implementation to provide better service facilities and a safer riding experience.

Based on the sources, the types and methods of road maintenance are categorized as follows:

The Department of Roads (DoR) in Nepal classifies maintenance activities into four primary categories:

- i. **Routine Maintenance:** These are frequent, year-round activities performed on all elements of the highway to ensure serviceability in all weather conditions. Examples include grass and bush cutting, cleaning drains and culverts, and repairing minor structural damage.
- ii. **Recurrent Maintenance:** These are localized activities carried out at regular intervals (typically six months to two years). They include sealing cracks, localized surface treatments, and repairing depressions and ruts.
- iii. **Periodic Maintenance:** Conducted at intervals of several years, these activities improve the overall functionality of the road. Scope includes renewing the wearing course, restoring road markings, and repainting metal bridges.
- iv. **Special Maintenance:** These are specialized or emergency works done to solve sudden problems like landslides or structural failures. Activities include removing debris, constructing temporary diversions, and strengthening pavement structures.

Maintenance can also be grouped based on its proactive or reactive nature:

- i. **Preventive Maintenance:** Actions taken to prevent premature deterioration and retard the progression of deficiencies to increase the pavement's useful life.
- ii. **Corrective Maintenance:** Actions taken to repair hazardous deficiencies and keep the highway within a tolerable level of serviceability.

b.

Highway Road maintenance is the task of preserving, repairing and restoring a system of roadways to keep the serviceable conditions of highway as normal as possible and best as practicable. It is importance as maintenance helps in providing better service facilities longer life and better appearance. It involves variety of planning, programming and scheduling to actual implementation in the field and monitoring.

The importance of highway maintenance is listed below:

- i. Highway maintenance ensures road safety after the repair of potholes, damaged shoulders, and reduced risk of road accidents.
- ii. It also extends the life of pavement by timely maintenance, preventing minor defects and cracks on it.
- iii. It also ensures smooth and continuous flow of traffic, reduces congestion and delay which maintains efficiency of the pavement.
- iv. Routine and preventive maintenance also reduces costly reconstruction and rehabilitation.
- v. Highway maintenance ensures effective drainage system as it comprises of cleaning and maintaining drains and culverts which prevents waterlogging.

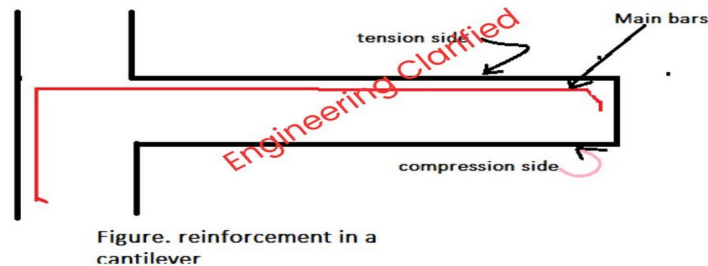
7. **What is the reason of providing reinforcement bars on the upper side of beam while designing cantilever beams? What is the purpose of designing double reinforced beams?**

Answer:

- In a cantilever beam, one end is fixed and the other is free.
- When such beam is subjected to downward loads, it bends such that the top fibers are in tension and bottom fibers are in compression.
- Since the concrete is weak in tension, steel reinforcements are placed at the top portion of beam to resist the tensile stresses.

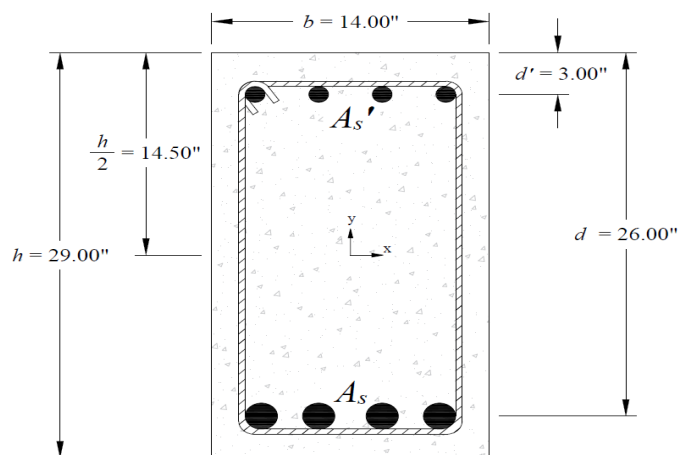
- The reinforcement prevents tensile cracking by increasing the flexural strength of beam.
- It also controls deflection and ensure the safe transfer of loads to the fixed support.

Why Reinforcement Is Provided In Top Of The Cantilever Beam



Purpose of designing doubly reinforced beams:

- To improve ductility and provide warning before collapse.
- To reduce long-term deflection in the beam due to creep and other external loads.
- To strengthen the structural performance under repeated loadings.
- To resist reversal of bending moments during seismic loadings or wind loadings.



8. **a. Discuss the various factors affecting permeability of soil.**
b. What are the causes of slope failures? Explain the use of bio-engineering in stabilizing slopes.

Answer:

a. Permeability of a soil is the property of soil that allows the water to pass through the voids present in it. It can be defined as the easiness with which water can flow through the pores of soil mass. Permeability property of a soil results in seepage ultimately causing water loss.

According to Darcy's law, the velocity of water is directly proportional to hydraulic gradient (i) in a pipe flow such that:

$$v = k \cdot I$$

where k = coefficient of permeability of soil or hydraulic conductivity.

The factors affecting permeability of soil are described below:

- i. **Void ratio:**

- Larger the void ratio, more porous the soil mass which increases the permeability of soil.
- $k \propto e^3 / (1+e)$

ii. Size of soil particles:

- Coarser the soil particles, larger the size of voids and higher permeability.
- According to Hazen's equation:

$$K = CD_{10}^2 \quad \text{where } C = \text{constant}$$

D_{10} = effective size of particle

iii. Shape of soil particles:

- Rounded soil particles have lower specific area and higher permeability while angular soil particles have high specific area and lesser permeability

iv. Viscosity of water:

- Viscous liquid reduces the permeability of soil.
- $k \propto 1/\mu$

v. Temperature:

- As the temperature increases, viscosity of liquid decreases which increases the permeability of soil.
- $k \propto 1/\eta$

b. If the shear strength of the soil is less than the shear stress developed, then the failure of slope takes place.

The causes of slope failure can be categorized into two main groups:

i. Increase in shear stress on the slope:

Causes:

- ✓ Heavy raining causes seepage along the slope, seepage pressure acts downward and shear stress increases by seepage pressure.
- ✓ Increase in load on the slope after construction
- ✓ Increase in inclination of slope by erosion
- ✓ Dynamic loading: certain component acts downward.

ii. Decrease in shear strength of soil on slope:

Causes:

- ✓ Heavy raining increases the water on slope, increase pore water pressure and reduces effective stress and reduces the shear strength.
- ✓ Weathering and rock on slope
- ✓ Loss of fines and soil on slope by erosion.
- ✓ Cyclic dynamic loading

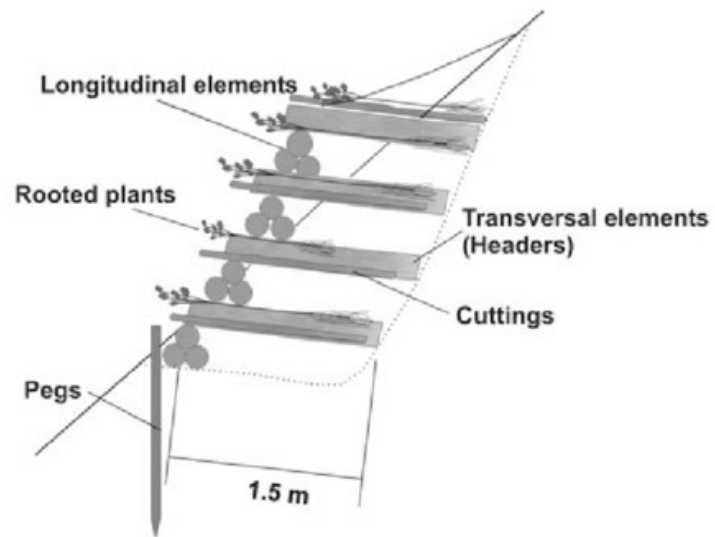
Stabilization of slope using vegetative systems with small civil engineering structures is known as bio-engineering. It combines biological measures (plants) with civil engineering works to improve slope stability in an environmentally friendly and cost-effective manner.

Principle of bio-engineering:

- The main concept of bio-engineering is the interaction between vegetative systems and small civil engineering structure.
- Plant roots bind soil particles and increase shear strength.
- Vegetation reduces the impact of rainfall and surface runoff.
- Plants absorb soil moisture, lowering pore water pressure.
- Natural cover protects slopes from erosion and weathering.

Example:

- Toe wall with bamboo plantation:



ii. Jute net with grass seedling:



Application of Bio-engineering helps to stabilize slopes, control erosion, and reduce landslide risks. It is widely used on hilly roads in Nepal because it is economical, environment-friendly and sustainable.

The different bio-engineering methods **along with its use in stabilizing slopes** are described below:

i. Turfing:

- Fast-growing grasses (e.g., Vetiver, Napier, Broom grass) are planted on exposed slopes.
- Dense root systems bind soil particles and reduce surface erosion caused by rainfall.
- Suitable for newly cut slopes and embankments.



ii. Shrubs and tree plantation:

- Deep-rooted shrubs and trees are planted to reinforce the soil.
- Roots increase soil shear strength and improve long-term slope stability.
- Also reduce runoff and improve the surrounding ecosystem.



iii. Fascines:

- Bundles of live branches are tied together and placed along contour lines.
- They slow runoff, trap sediments, and encourage vegetation growth.
- Commonly used on moderately steep slopes.



iv. Check dams:

- Small barriers made from bamboo, brushwood, or live stakes are built across gullies.
- Reduce runoff velocity, trap sediments, and prevent gully erosion.
- Particularly useful in drainage channels.



v. Palisades:

- Wooden stakes with live cuttings are driven into unstable slopes.
- Provide immediate support while developing a living root network.
- Suitable for shallow failures and eroded slopes.



vi. Jute netting:

- Natural fiber mats made of coconut (coir) or jute are spread over exposed soil.
- Protect the soil from rain splash and erosion until vegetation becomes established.
- Eventually decompose naturally.



vii. Gabion walls and crib walls:

- Gabion or timber crib walls are filled with soil and planted with grasses or shrubs.
- Combine structural support with biological stabilization.
- Suitable for high cut slopes and retaining structures.

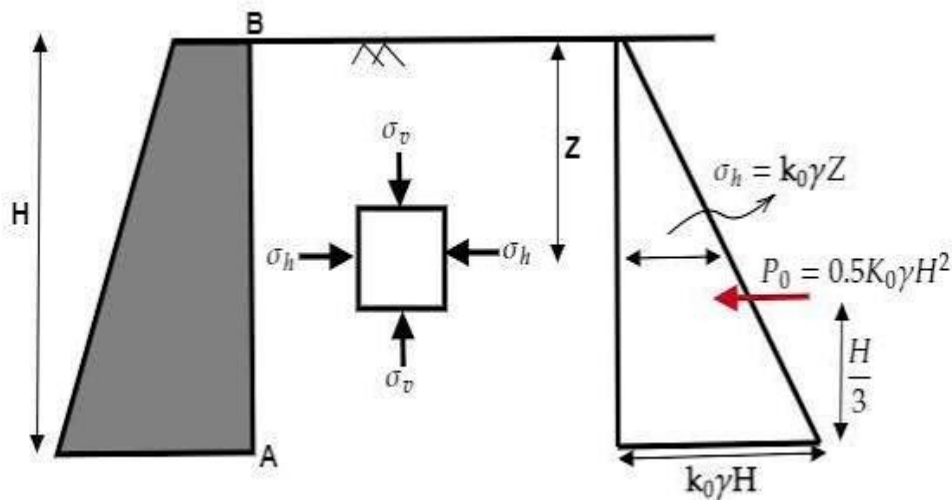
9. **a. Write the assumption of Rankine's theory and derive the expression for active and passive pressures.**

Answer:

The expression for active earth pressure on the vertical back face of earthwork supporting granular soil having horizontal surface is given by Rankine's theory.

The assumptions for Rankine's theory:

- The back of the retaining wall is vertical and smooth (no wall friction).
- The backfill is dry, homogeneous, cohesionless (granular) soil.
- The ground surface is horizontal.
- The soil is in the plastic equilibrium state.
- Failure occurs along a plane surface.



Let us consider a soil element at a depth h below the ground surface.

Vertical stress due to overburden:

$$\sigma_v = \gamma h$$

where:

$$\gamma = \text{unit weight of soil (kN/m}^3\text{)}$$

$$h = \text{depth below ground level (m)}$$

The horizontal stress under active condition is

$$\sigma_h = K_a \sigma_v$$

where K_a is the coefficient of active earth pressure.

According to Rankine's failure criterion,

$$K_a = \frac{1 - \sin \phi}{1 + \sin \phi}$$

or

$$K_a = \tan^2 \left(45^\circ - \frac{\phi}{2} \right)$$

where

ϕ = angle of internal friction of soil.

Substituting

$$\sigma_h = K_a \gamma h$$

For a retaining wall of height H ,

Pressure at the top:

$$p = 0$$

Pressure at the bottom:

$$p = K_a \gamma H$$

The total active earth pressure equals the area of the pressure triangle.

$$P_a = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$P_a = \frac{1}{2} (K_a \gamma H) (H)$$

Therefore,

$$P_a = \frac{1}{2} K_a \gamma H^2$$

This force acts horizontally on the wall.

Since the pressure distribution is triangular, the resultant acts at the centroid of the triangle.

Hence,

$$\text{Point of application} = \frac{H}{3} \text{ above the base}$$

b. For a strip footing on cohesive soil ($\phi=0$), Terzaghi's bearing capacity equation is:

$$q_u = c N_c + \gamma D_f N_q + \frac{1}{2} \gamma B N_\gamma$$

For cohesive soil ($\phi=0$):

$$N_c = 5.7$$

$$N_q = 1$$

$$N_\gamma = 0$$

Therefore,

$$q_u = 5.7 c + \gamma D_f$$

where:

q_u = ultimate bearing capacity

c = cohesion of soil

γ = unit weight of soil

D_f = depth of foundation

B = width (breadth) of footing

10. a. What types of foundations are used in bridge construction? Explain the each in brief.

b. How do you determine the bearing capacity of soils? What are the factors influencing bearing capacity?

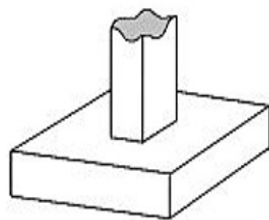
Answer:

a. In bridge construction, foundations are structural elements designed to receive loads from the superstructure and transfer them safely to stable soil or rock layers. These foundations are broadly classified into:

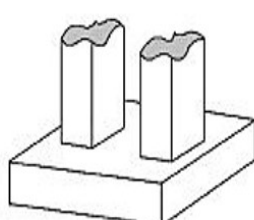
Shallow Foundations and Deep Foundations.

i. **Shallow foundations:**

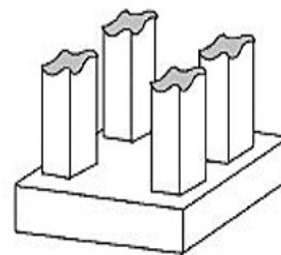
- These are used when firm soil or sound rock with adequate bearing capacity is available near the ground surface.
- They are typically defined as foundations where the depth is less than or equal to the width.
- Suitable for smaller bridges, residential-scale crossings, or bridges in hilly regions where foundations can be placed directly on sound rock.
- Used when the soil with adequate bearing capacity is available near the ground surface.



Isolated Footing



Continuous footing



Mat footing

ii. **Deep foundations:**

Deep foundations are required when the upper soil layers are weak, compressible, or susceptible to erosion (scour), necessitating the transfer of loads to stronger strata at greater depths.

• **Pile Foundations:**

- ✓ Consists of long, slender members that are either driven or bored into the ground.
- ✓ They transmit loads to deeper soil strata through end bearing (resting on a hard layer), skin friction (adhesion along the pile sides), or both.
- ✓ Preferred for bridges when the bearing capacity of surface soil is low or for heavy structures in waterlogged areas.

• **Well Foundations (Caissons):**

- ✓ These are large, watertight structures that are gradually sunk into the ground by excavating material from within them.
- ✓ Load is transferred through base resistance and side resistance after the well reaches its required depth.
- ✓ Specifically preferred for bridge piers in rivers as they provide strong resistance to heavy lateral forces and can be sunk below the scour depth to ensure stability during floods.

- **Pier Foundations:**

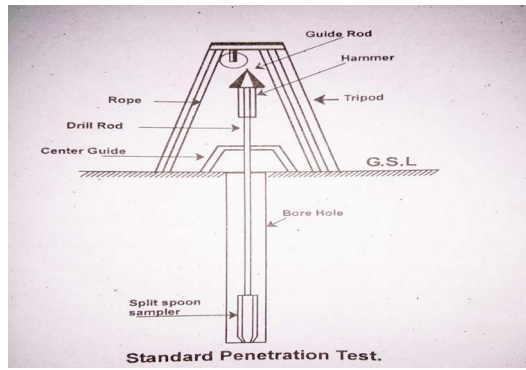
- ✓ These are vertical columns similar to piles but typically larger in diameter, used to transfer structural loads to a hard stratum at depth.

The choice of foundation depends on several critical factors:

- i. **Geological Conditions:**
 - Foundations are placed on firm, non-erodible strata whenever possible to ensure reliability.
 - ii. **Scour Depth:**
 - For river bridges, the foundation must be fixed below the calculated maximum scour depth to prevent the structure from being undermined during high-velocity flood events.
 - iii. **Bridge Type:**
 - Heavier structures like arch bridges require very strong foundations, whereas lighter steel truss or cable-supported bridges may be feasible on slightly weaker strata.
- b.
- Bearing capacity of soil is defined as maximum load per unit area that soil mass can safely withstand without undergoing shear failure or excessive settlement.
 - It is the ability of a soil mass to safely support structural loads without any failure.
 - It is determined through field tests, laboratory tests, and theoretical analysis.
 - Methods of determining the bearing capacity of soil:
 - i. **Field tests:**

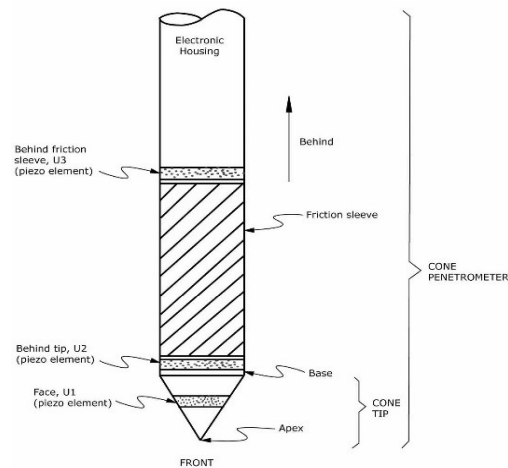
Bearing capacity of soil is determined directly at site.

 - a. **Plate load test:**
 - A steel plate is placed at the foundation level.
 - Load is applied gradually using a hydraulic jack.
 - Settlement is measured for each load increment.
 - The load-settlement curve is plotted, and the ultimate bearing capacity is determined.
 - Suitable for shallow foundations.
 - b. **Standard penetration test:**
 - SPT tests is one of the most commonly used field tests.
 - The test is conducted in a borehole using a split-spoon sampler from where N-value is determined.
 - The number of blows required to drive the sampler 300 mm is called SPT N-value.



c. Cone penetration test:

- In cone penetration test, a cone is pushed into ground at a constant rate of 2 cm per second.
- It measures cone resistance and sleeve friction.
- The number of blows of hammer required to penetrate a cone to 300 mm depth is called CPT N_c value.



ii. Lab tests:

- Bearing capacity of the soil is determined using the soil samples collected from the field.
- Shear strength tests such as direct shear test, triaxial compression test and unconfined compression tests are carried out in laboratory.
- These tests help to determine cohesion, angle of internal friction and unit weight, used in the bearing capacity equations.

iii. Theoretical analysis:

Bearing capacity can be determined by using Terzaghi's bearing capacity equations.

$$q_u = c N_c + \gamma D_f N_q + 0.5 \gamma B N_\gamma$$

Where :

q_u = Ultimate bearing capacity

c = Cohesion of soil

γ = Unit weight of soil

$D_f = \text{Depth of foundation}$

$B = \text{Width of footing}$

$N_c, N_q, N_\gamma = \text{Bearing capacity factors depending on } \phi$

The safe bearing capacity is obtained by applying a factor of safety (usually 2.5–3):

$$\text{SBC} = \frac{q_u}{\text{Factor of Safety}}$$

Factors affecting bearing capacity of soil:

i. Size of footing:

- Bearing capacity of the soil is directly proportional to the width and depth of footing.

$$q \propto B$$

$$q \propto D_f$$

ii. Shape of footing:

- Terzaghi gave a bearing capacity formula for strip footing having infinite length.

$$q_u = c N_c + \gamma D_f N_q + 0.5 \gamma B N_\gamma$$

- Hence, he modified the theory for finite length of strip footing by introducing shape factors.
- Shape of the footing affects on shape and size of the rigid cone formed which affects on cohesive force and weight of rigid cone.

$$q_u = c N_c S_c + \gamma D_f N_q S_q + 0.5 \gamma B N_\gamma S_\gamma$$

iii. Type of soil:

For cohesive soil such as clay, bearing capacity of soil depends only upon cohesion.

For cohesionless soil such as sand, bearing capacity of soil depends upon angle of friction.

iv. Position of GWT:

Groundwater table makes the soil mass submerged. So, unit weight of the soil changes into submerged unit weight.

$$\gamma' = \frac{\gamma}{2}$$

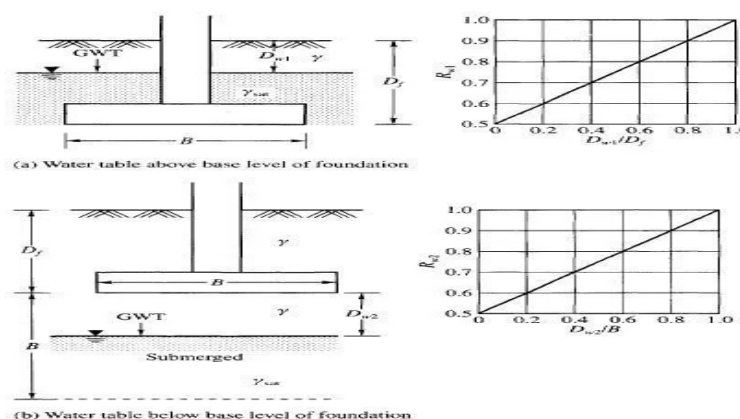


Figure 12.10 Effect of WT on bearing capacity

When Groundwater Table is at Foundation Level:

The soil below the foundation becomes submerged. Therefore, replace γ in the third term with the submerged unit weight γ' :

$$\gamma' = \gamma_{sat} - \gamma_w$$

Thus,

$$q_u = c N_c + \gamma D_f N_q + 0.5 \gamma' B N_\gamma$$

When Groundwater Table is at Ground Surface:

The soil above and below the foundation is submerged. Use γ' in both second and third terms:

$$q_u = c N_c + \gamma' D_f N_q + 0.5 \gamma' B N_\gamma$$

Since: $\gamma' = \gamma_{sat} - \gamma_w < \gamma$, a rise in the groundwater table decreases the effective unit weight of soil, causing the ultimate bearing capacity q_u to decrease.

v. Principle mode of shear failure:

- Terzaghi's theory was derived with an assumption for general shear failure.
- He modified this theory for local shear failure.
- In comparison to general shear failure, mobilized shear strength of soil in local shear failure is less.

$$C_m = \frac{2}{3} C$$